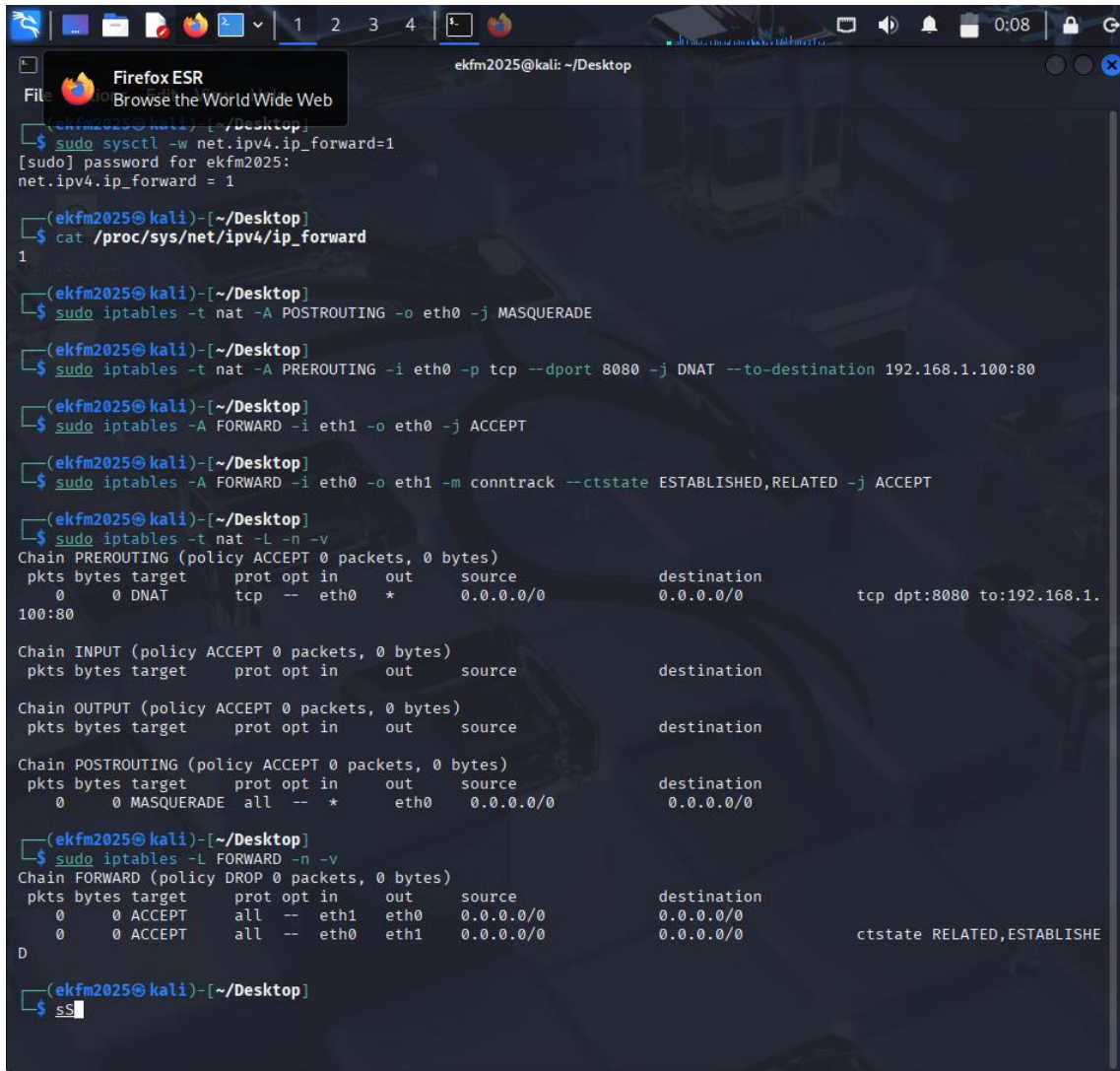


LAB ACTIVITY: Advanced Firewall Routing and NAT



```
ekfm2025@kali: ~/Desktop
$ sudo sysctl -w net.ipv4.ip_forward=1
[sudo] password for ekfm2025:
net.ipv4.ip_forward = 1

(ekfm2025@kali)-[~/Desktop]
$ cat /proc/sys/net/ipv4/ip_forward
1

(ekfm2025@kali)-[~/Desktop]
$ sudo iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE

(ekfm2025@kali)-[~/Desktop]
$ sudo iptables -t nat -A PREROUTING -i eth0 -p tcp --dport 8080 -j DNAT --to-destination 192.168.1.100:80

(ekfm2025@kali)-[~/Desktop]
$ sudo iptables -A FORWARD -i eth1 -o eth0 -j ACCEPT

(ekfm2025@kali)-[~/Desktop]
$ sudo iptables -A FORWARD -i eth0 -o eth1 -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT

(ekfm2025@kali)-[~/Desktop]
$ sudo iptables -t nat -L -n -v
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out    source        destination
0      0 DNAT          tcp  --  eth0   *      0.0.0.0/0     0.0.0.0/0     tcp dpt:8080 to:192.168.1.100:80

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out    source        destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out    source        destination

Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out    source        destination
0      0 MASQUERADE    all  --  *      eth0   0.0.0.0/0     0.0.0.0/0

(ekfm2025@kali)-[~/Desktop]
$ sudo iptables -L FORWARD -n -v
Chain FORWARD (policy DROP 0 packets, 0 bytes)
pkts bytes target      prot opt in     out    source        destination
0      0 ACCEPT      all  --  eth1   eth0   0.0.0.0/0     0.0.0.0/0
0      0 ACCEPT      all  --  eth0   eth1   0.0.0.0/0     0.0.0.0/0     ctstate RELATED,ESTABLISHED
```

1. Obfuscation: How does MASQUERADE protect the Confidentiality of the internal network's size and structure?

- By using MASQUERADE, all outgoing traffic appears to originate from the Gateway's single public IP. This hides the internal network's structure and the exact number of devices from outside observers.

2. Port Remapping: We mapped Public Port 8080 to Internal Port 80. What is the security advantage of using a non-standard port on the public-facing interface?

- By mapping internal port 80 to public port 8080, a layer of security is created through obscurity. The majority of automated scripts look for port 80; we can lessen the

amount of "noise" and automated attacks that target your web server by using a high-port.

3. The "Missing Link": If you successfully configure Part 2 (SNAT) but forget Part 1 (IP Forwarding), will the internal clients have internet? Why or why not?

- If we had skipped the `sysctl` step (Part 1), the NAT rules would still exist in the tables, but the Linux kernel would simply drop the packets instead of passing them between `eth1` and `eth0`.